METRIC

MIL-PRF-32833
13 Ocotber 2026

PERFORMANCE SPECIFICATION

HEAT EXCHANGERS, PLATE TYPE, NAVAL SHIPBOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers plate type heat exchangers (PHE) for use with seawater, fuel oil, lubricating oil, hydraulic oil, freshwater, and electronic cooling water for naval shipboard applications.

1.2 Classification. The coolers are of the following classes as specified (see 6.2):

a. Class 1 - Gasketed plate heat exchangers.

b. Class 2 - Welded/brazed/fusion-bounded plate heat exchangers.

c. Class 3 - Plate and shell heat exchangers.

d. Class 4 - Welded block plate heat exchangers.

1.3 Part or identifying number (PIN). The PIN to be used for PHEs acquired to this specification is constructed as follows:

M	**XXXXX**	**-**	**X**	**X**	**X**	**XXXXX**	**X**	**XXXXX**	**CXX**	**DXX**
Prefix for Military Specification	Specification Number	-	Class (see 1.2)	Cooling media (see 1.3.1.1)	Cooled media (see 1.3.1.2)	Test pressure, kilopascals (kPa)	Plate Material (see 1.3.1.3)	Design heat transfer, kilowatt (kW)	Design flow rate, cooling media, cubic meters/min (m^3/min)	Design flow rate, cooled media, m^3/min

1.3.1 Media PIN codes.

1.3.1.1 Cooling media PIN codes.

a. Seawater.

b. Fresh water (non-potable) from the ship's reverse osmosis (RO) or distilling plants.

c. Fresh water from external sources such as lakes or shore-supplied in ports.

d. Chilled water from the ship's chill water system.

e. Deionized fresh water from the ship's system.

f. Other (as specified [see 6.2]).

1.3.1.2 Cooled media PIN codes.

a. Electronic cooling water.

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b. Glycol-based coolants.

c. Fuel oil in accordance with MIL-DTL-5624 (JP-5) or MIL-DTL-16884 (naval distillate fuel).

d. Lubricating oil in accordance with MIL-PRF-9000, MIL-PRF-17331, or MIL-PRF-23699.

e. Hydraulic oil in accordance with MIL-PRF-17111, MIL-DTL-32353, or MIL-PRF-46170.

f. Steam.

g. Other (as specified[see 6.2]).

1.3.1.3 Plate material PIN codes. See table I.

a. Copper-nickel

b. Bronze

c. Nickel-aluminum

d. Corrosion-resistant steel (CRES)[1]

e. Titanium

f. Carbon steel

g. Stainless steel

h. Nickel-molybdenum-chromium

i. Other (as specified [see 6.2])

1.3.2 PIN example. An example of a PHE PIN is M32833-2AC00670E00020C10D10. This is a class 2, seawater cooled, fuel oil cooling, titanium plate, 670 kPa (100 pounds per square inch [psi]), 20 kW, 10 cubic meter cooling flow rate, 10 cubic meter cooled flow rate, plate heat exchanger.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52401	-	Bearing, Sleeve (Steel-Backed)
A-A-59782	-	Heat Exchanger Cleaning Compound

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-1222	-	Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for
MIL-PRF-2818	-	Insulation Blanket, Thermal
MIL-DTL-5624	-	Turbine Fuel, Aviation, Grade JP-5 (NATO F-44)

[1] CRES is a subset of stainless steel, with at least 14 percent chromium, while stainless steel contains at least 10.5 percent chromium.

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MIL-PRF-9000 - Lubricating Oil, Shipboard Internal Combustion Engine, High-Output Diesel
MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for
MIL-P-15024/5 - Plate, Identification
MIL-C-15726 - Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod and Wire
MIL-DTL-16884 - Fuel, Naval Distillate
MIL-PRF-17111 - Fluid, Power Transmission
MIL-PRF-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-R-21252 - Rubber Sheet, Solid, Synthetic, Shipboard Water Evaporator Gasketing
MIL-G-21610 - Gaskets, Heat Exchanger, Various Cross Section Rings, Synthetic Rubber
MIL-DTL-22050 - Gasket and Packing Material, Rubber, for Use with Polar Fluids, Steam, and Air at Moderately High Temperatures
MIL-S-22698 - Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural
MIL-PRF-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Numbers: O-152, O-154, O-156, and O-167
MIL-DTL-24441 - Paint, Epoxy-Polyamide, General Specification for
MIL-DTL-24441/20 - Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type III
MIL-DTL-24441/21 - Paint, Epoxy-Polyamide, Haze Gray Formula 151, Type III
MIL-B-24480 - Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service
MIL-DTL-24696 - Gasket, Sheet, Non-Asbestos
MIL-DTL-32353 - Hydraulic and Lubricating Oil, Synthetic Hydrocarbon Base
MIL-PRF-46170 - Hydraulic Fluid, Rust Inhibited, Fire Resistant, Synthetic Hydrocarbon Base, NATO Code No. H-544

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships
MIL-STD-889 - Galvanic Compatibility of Electrically Conductive Materials

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment
MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume 1

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this publication are available online at https://www.gsaglobalsupply.gsa.gov.)

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

505-8289963 - Titanium Seawater Piping, Electrical Isolation to Prevent Galvanic Corrosion

804-5284201 - Reinforced Rubber, Gasket Material, for Pipeline Joints

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

S9505-A1-DDT-010 - NAVSEA Design Practices and Criteria Manual for Titanium Piping Systems

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of these publications are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). These publications are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Material Target List

(Copies of this publication are available online at www.aia-aerospace.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

NACE Standard SP0286 - Electrical Isolation of Cathodically Protected Pipelines

SSPC-SP 10/NACE No. 2 - Near-White Metal Blast Cleaning

(Copies of these publications are available online at https://webstore.ansi.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24, Metric/Inch Standard

ASME BPVC.VIII.1 - BPVC Section VIII - Rules for Construction of Pressure Vessels Division I

(Copies of these publications are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A53/A53M - Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A106/A106M - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service

ASTM A131/A131M - Standard Specification for Structural Steel for Ships

ASTM A193/193M - Standard Specification for Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

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ASTM A516/A516M	-	Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service
ASTM A572/A572M	-	Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel
ASTM B265	-	Standard Specification for Titanium and Titanium Alloy Strip, Sheet, and Plate
ASTM B369/B369M	-	Standard Specification for Copper-Nickel Alloy Castings
ASTM B381	-	Standard Specification for Titanium and Titanium Alloy Forgings
ASTM B466/B466M	-	Standard Specification for Seamless Copper-Nickel Pipe and Tube
ASTM B575	-	Standard Specification for Low-Carbon Nickel-Chromium-Molybdenum, Low-Carbon Nickel-Chromium-Molybdenum-Copper, Low-Carbon Nickel-Chromium-Molybdenum-Tantalum, Low-Carbon Nickel-Chromium-Molybdenum-Tungsten, and Low-Carbon Nickel-Molybdenum-Chromium
ASTM D3294	-	Standard Specification for Polytetrafluoroethylene (PTFE) Resin Molded Sheet and Molded Basic Shapes

(Copies of these publications are available online at www.astm.org.)

INTERNATIONAL SPECIFICATIONS ORGANIZATION (ISO)

ISO 3506-1	-	Fasteners – Mechanical properties of corrosion-resistant stainless steel fasteners – Part 1: Bolts, screws and studs with specified grades and property classes
ISO 15547-1	-	Petroleum, petrochemical and natural gas industries – Plate-type heat exchangers – Part 1: Plate-and-frame heat exchangers
ISO 15547-2	-	Petroleum, petrochemical and natural gas industries – Plate-type heat exchangers – Part 2: Brazed aluminum plate-fin heat exchangers

(Copies of these publications are available online at www.iso.org.)

SAE INTERNATIONAL

SAE AMS7259	-	Rubber: Fluorocarbon (FKM) High Temperature/Fluid Resistant Low Compression Set/85 to 95 Hardness for Seals in Fuel Systems and Specific Engine Oil Systems
SAE AMS7276	-	Rubber: Fluorocarbon (FKM) High Temperature – Fluid Resistant Low Compression Set/70 to 80 Type A Hardness for Seals in Fuel Systems and Specific Engine Oil Systems
SAE AS8943	-	Bearings, Sleeve, Plain and Flanged, Self-Lubricating -65 to +250 °F
SAE HS-1086	-	Metals and Alloys in the Unified Numbering System

(Copies of these publications are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Materials. Materials for plate heat exchangers (PHE) assemblies shall be corrosion-resistant, or materials protected against corrosion after fabrication. Material degraded during the fabrication process shall be normalized to restore those properties before being assembled in any PHE assembly. Selected materials shall be capable of

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meeting all the operational and environmental requirements specified herein and shall be compatible with the specified (see 6.2) ship service. The materials listed in table I are recommended; however, materials having equal or better physical and chemical properties than the materials specified therein may be utilized with approval from the Technical Authority (see 6.4).

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TABLE I. Recommended heat exchanger materials.[1/]

Component	Example materials	Typical reference	Applicable services
Frames, carry and guide bars, supports	Steel, grade A, class U	MIL-S-22698	All
	Steel, grade B, seamless	ASTM A131/A131M	All
	Commercial steels	ASTM A36, ASTM A516/A516M, ASTM A572/A572M	All
	CRES UNS[2/] S30400, S30403, S31254, S31600, S31603	-	All
Tie rods, fasteners, bolts, or studs	Carbon steel	Commercial	All
	Alloy steel	Commercial	All
	Nickel-copper, grade 400 (UNS N04400), nickel copper-aluminum, grade 500 (UNS N05500)	MIL-DTL-1222	All
	CRES (UNS S30400, S30403, S31254, S31600, S31603)	ISO 3506-1, grade A4, ASTM A193/A193M, grade B8M	All
Plates	Copper-nickel-alloy 70-30 (UNS C71500)	MIL-C-15726	Freshwater, seawater, glycol-based coolants, hydraulic oil
	Copper-nickel-alloy 90-10 (UNS C70600)	ASTM B369/B369M	Freshwater, seawater, glycol-based coolants, hydraulic oil
	Bronzes	Commercial, as determined for specific application	
	Bronze, nickel-aluminum (UNS C95800)	MIL-B-24480	
	CRES UNS[2/] S30400, S30403, S31254, S31600, S31603	ASTM A240/A240M	Freshwater, deionized water, glycol-based coolants, fuel oil, lube oil, hydraulic oil
	Titanium[3/] (UNS R50250, R50400)	ASTM B265, ASTM B381	Freshwater, seawater, fuel oil, lube oil, hydraulic oil
	Carbon steels	Commercial, as determined for specific application	All
	Stainless steels	Commercial, as determined for specific application	All
	Nickel-molybdenum-chromium alloy (UNS N10276)	ASTM B575	Freshwater, seawater, glycol-based coolants, hydraulic oil

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TABLE I. Recommended heat exchanger materials – Continued.

Component	Example materials	Typical reference	Applicable services
Piping (inlet and outlet)	Steel	ASTM A53/A53M ASTM A106/A106M	Freshwater, glycol-based coolants, fuel oil, lube oil, hydraulic oil
	CRES	-	Freshwater, deionized water, glycol-based coolants, fuel oil, lube oil, hydraulic oil
	Copper-nickel-alloy	ASTM B466/B466M	Freshwater, seawater, glycol-based coolants, hydraulic oil
Plate gaskets[4/] (hot side/ cold side)	Fluorocarbon rubber	SAE AMS7276	Freshwater, seawater, chlorinated seawater, lube oil, hydraulic oil
		SAE AMS7259	Freshwater
	Synthetic rubber	MIL-G-21610	Freshwater, seawater, fuel oil, ethylene glycol cooling water
		MIL-R-21252	Seawater
	Ethylene propylene diene monomer (EPDM)	MIL-DTL-22050	Freshwater
	Non-asbestos sheet	MIL-DTL-24696	Steam, freshwater, seawater, fuel oil, hydraulic oil, JP-5, ethylene glycol cooling water
	Composite sheet	Garlock ® (CAGE 73680) 9900, or equal	Freshwater, seawater, fuel oil, hydraulic oil, JP-5
	Nylon inserted rubber	804-5284201, types 1, 2, or 4	Freshwater, seawater, chlorinated seawater, ethylene glycol cooling water
	(PTFE	ASTM D3294	Freshwater, seawater, fuel oil, hydraulic oil, JP-5, ethylene glycol cooling water
Frame gaskets	Non-asbestos sheet[5/]	MIL-DTL-24696	Steam, water, lube and hydraulic oils
Sacrificial liners[6/]	As determined by intended service use	As determined by intended service use	-
Shrouds	CRES UNS[2/] S30400, S30403, S31254, S31600, S31603	-	All
	Aluminum 5XXX alloys	ASTM B209/B209	All

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TABLE I. Recommended heat exchanger materials – Continued.

NOTES:

1/ Equivalent materials may be substituted as approved by the NAVSEA Technical Authority (see 6.4). Gasket materials for inlet and outlet piping connections shall be in accordance with ship specifications or MIL-STD-777, as applicable.

2/ Unified Numbering System (UNS) in accordance with SAE HS-1086.

3/ When titanium plates are used, all effort shall be made to minimize galvanic corrosion. While applicable only to titanium seawater piping systems, S9505-A1-DDT-010 and 505-8289963 shall be used for guidance on titanium galvanic corrosion mitigation methods. NACE Standard SP0286 shall also be used for guidance. If any conflicts between the application of the NACE standard and the requirements of these specifications arise, the requirements of these specifications shall take precedence. If isolation in accordance with the above references is not practical for PHE plates, alternative galvanic corrosion countermeasures shall be submitted for approval to the NAVSEA Technical Authority (see 6.4). Isolation materials selected shall be compatible with the intended pressure and service and shall not present fire, safety, or other hazards.

4/ Gasket material only usable for heat exchanger plates.

5/ Gasket material only usable for heat exchanger frame, covers, or bolted plates.

6/ If utilized in specific design.

3.2.1 Prohibited materials. Exchangers shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.2.2 Welding and allied processes. Fabrication welding and inspection shall be in accordance with S9074-AR-GIB-010/278, except as modified herein. Welded joints shall be ultrasonically testable except for vent and drain nipples. Vent, drain nipple, and instrumentation welded connections shall meet the 100 percent weld efficiency requirements for pressure vessels.

3.2.2.1 Casting inspection and repair. Casting inspection and repair shall be in accordance with S9074-AR-GIB-010/278.

3.2.2.2 Brazing. Brazing shall be in accordance with 0900-LP-001-7000. Brazed joints shall be ultrasonically inspected.

3.2.3 Dissimilar metals. Coupled dissimilar metals shall be protected against corrosion in a marine environment in accordance with MIL-STD-889.

3.2.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Environmental considerations.

3.3.1 Shock resistance. Unless otherwise specified (see 6.2), all PHEs shall withstand the high-impact shock test as specified for grade A, class 1 equipment of MIL-DTL-901, with no leakage or degradation of heat transfer capacity.

3.3.2 Mechanical vibration. Operating plate heat exchangers shall conform to the vibration requirements of MIL-STD-167-1, type I.

3.3.3 Operational environment. Unless otherwise specified (see 6.2), PHEs can be expected to operate in internal shipboard spaces. When seawater is utilized, the PHEs shall be fully operational with water temperatures from -2 °C (28.4 °F) to 38 °C (100 °F).

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3.4 Design. A plate heat exchanger consists of parallel plates separated by gaskets, plates welded or brazed together, or plates assembled in blocks (see figures 1 through 6). The structural design for PHEs shall be in accordance with ASME BPVC.VIII.1. Basic designs shall be in accordance with ISO 15547-1 or ISO 15547-2, as applicable.

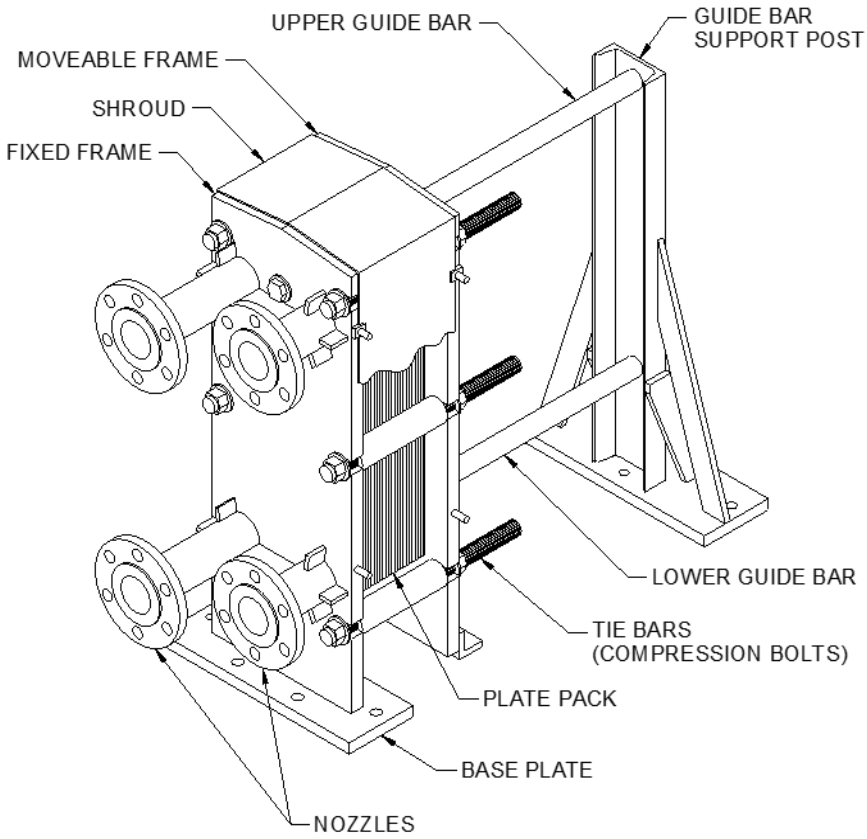


FIGURE 1. Typical gasketed plate PHE.

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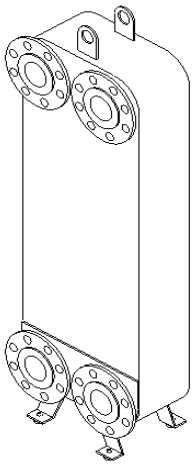


FIGURE 2. Typical welded plate PHE.

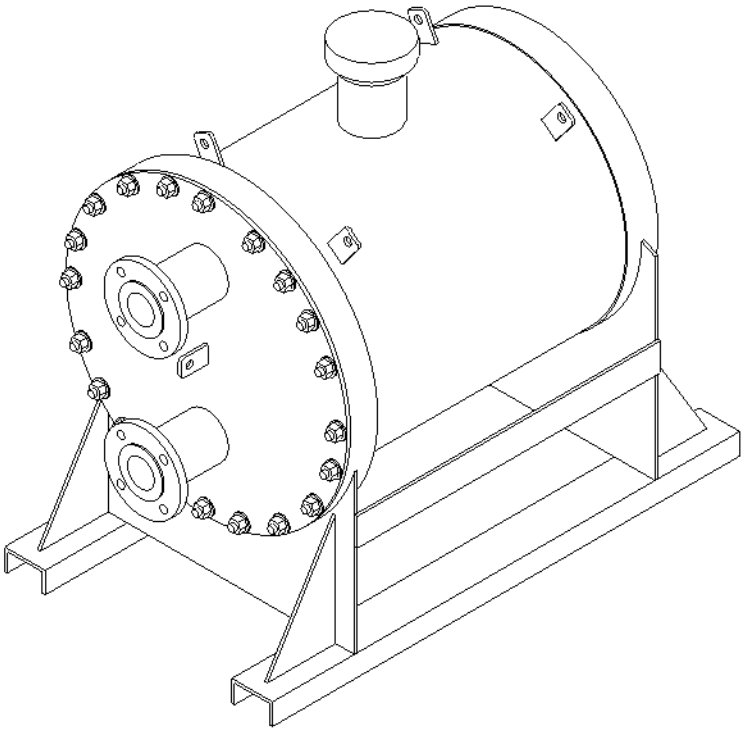


FIGURE 3. Typical plate-and-shell PHE.

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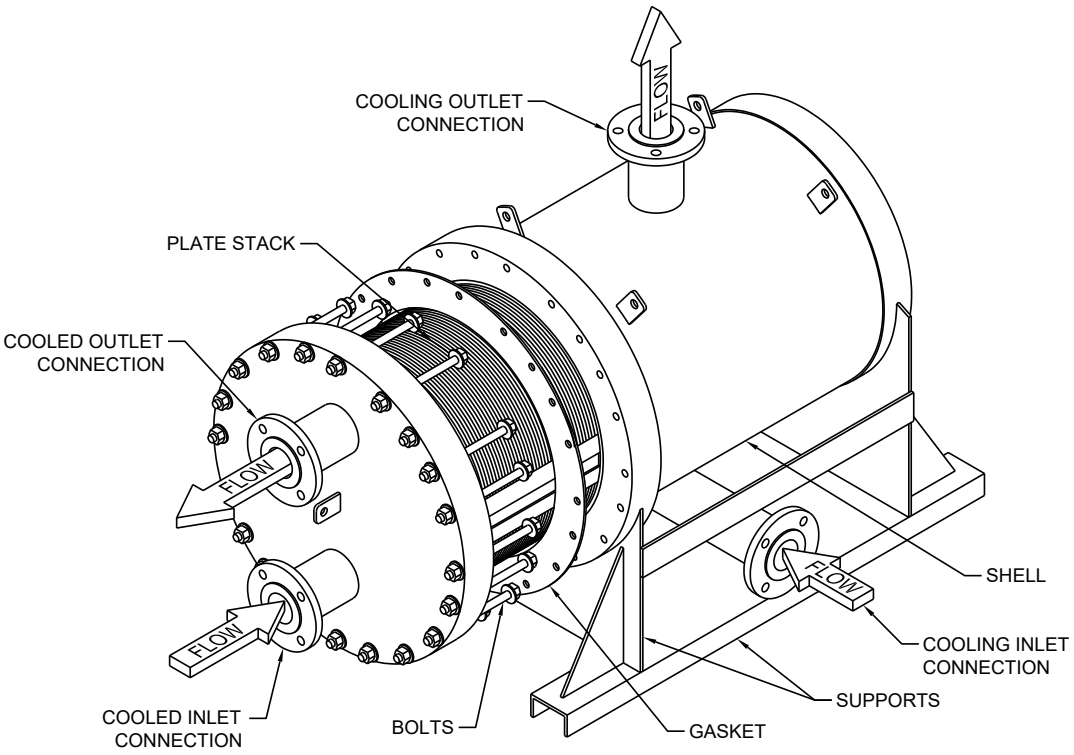


FIGURE 4. Exploded view of typical plate-and-shell PHE.

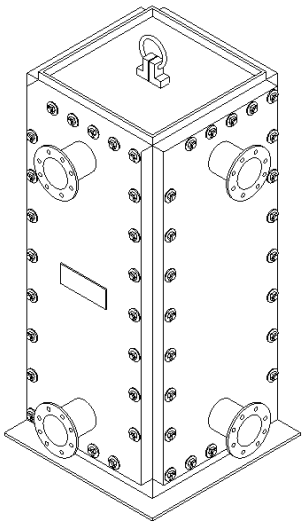


FIGURE 5. Typical welded block PHE.

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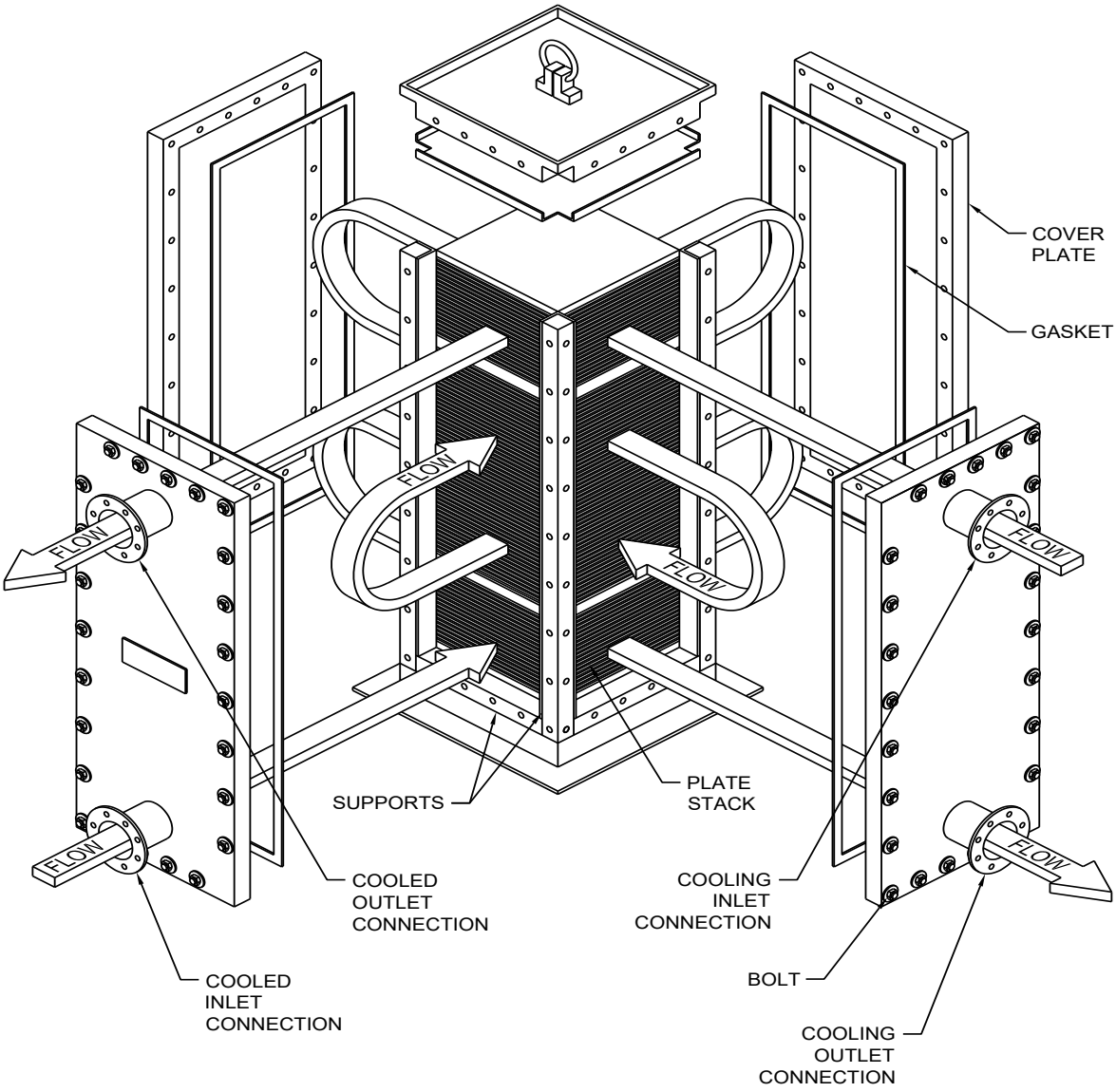


FIGURE 6. Exploded view of typical welded block PHE.

3.4.1 Design philosophy. The design philosophy for the PHEs is to achieve the performance, safety, maintainability, and reliability specified herein. The weight of the unit shall be kept to a minimum while meeting the requirements herein and as otherwise specified (see 6.2). The units shall be designed to reduce maintenance requirements. Additionally, PHEs shall:

a. Have heat transfer surfaces that are easily accessible for inspection or mechanical cleaning, allowing these operations to be performed by ship's force underway without the need for depot support. Gasketed PHE shall include provisions for chemical cleaning and be compatible with the heat exchanger cleaning products of A-A-59782.

b. Be designed such that the frame and tie bolts of the gasketed PHE (class 1) shall permit the future installation of at least 15 percent additional plates. To ensure shock compliance, class 1 PHE shock testing shall be performed with the additional plates installed.

c. Allow individual replacement of gasketed plates and pairs of semi-welded plates without requiring complete disassembly of the plate heat exchanger (for classes 1 and 4).

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d. Incorporate means for correct alignment of the plates and gaskets without use of special tools or techniques. Special tools are defined as tools not listed in the GSA Global Supply Catalog.

3.4.2 Working fluids. PHE working fluids shall be as specified (see 6.2). Typically, the cooling media will be one of the following:

a. Seawater.

b. Fresh water (non-potable) from the ship's RO or distilling plants.

c. Fresh water from external sources such as lakes or shore-supplied in ports.

d. Chilled water from the ship's chill water system.

e. Deionized fresh water from the ship's system.

3.4.2.1 Cooled media. The working fluid being cooled will typically be:

a. Electronic cooling water.

b. Glycol based coolants.

c. Fuel oil in accordance with MIL-DTL-5624 (JP-5) or MIL-DTL-16884 (naval distillate fuel).

d. Lubricating oil in accordance with MIL-PRF-9000, MIL-PRF-17331, or MIL-PRF-23699.

e. Hydraulic oil in accordance with MIL-PRF-17111, MIL-DTL-32353, or MIL-PRF-46170.

f. Steam.

3.5 Construction.

3.5.1 Frames. Frames shall provide the strength and rigidity necessary for the PHE to function properly.

3.5.1.1 Gasketed or welded plate HPE, class 1.

3.5.1.1.1 Fixed frame. The fixed frame, which is at the front of the unit, shall be bolted to the deck, equipment skid, or other foundation structure. All inlet and outlet connections shall be located in the fixed frame, such that the movable frame and plates can be removed without breaking pipe joints, giving easy access to the plates for cleaning and repair.

3.5.1.1.2 Movable frame and guide bars. The movable frame shall provide support at the back of the plate pack. As the tie bars are tightened, the movable frame shall be drawn toward the fixed frame, compressing the plate pack. The movable frame shall slide back and forth on upper and lower guide bars. Guide bars shall be long enough for access to the entire plate pack. When the movable frame has been pulled up to the proper plate pack dimension, support foot shall be provided to bolt to the foundation.

3.5.1.1.3 Tie bars. The tie bars (also called compression, tightening, or tie bolts) shall connect the movable frame to the fixed frame. When they are tightened, they shall compress the plate pack between the two frames. The specific sequence shall be defined to ensure plate pack compression and to prevent plate and gasket damage. Tie bar bearings, if employed, shall be sleeve or journal bearings in accordance with A-A-52401 or SAE AS8943.

3.5.1.1.4 Plates. The plates shall be pressed from single sheets of metal. They shall be sized according to the required heat transfer and fluids utilized. Plate chevron design shall maximize turbulent flow to reduce fouling. Plates shall prevent fluid intermixing and provide a visual indication of leaks into a drip pan.

3.5.1.1.5 Gaskets. For gasketed PHEs (class 1), each plate shall have a gasket that runs around its edge and circles either the hot or the cold fluid inlet and outlet ports. Grooves shall ensure that the gaskets are positioned correctly and are held in place when the unit is under pressure. Gaskets shall be formed from an elastomeric material designed to retain its sealing properties at the operating temperature and pressure. All gaskets for a given PHE unit shall be the same design and material.

3.5.1.1.6 Shroud. The shroud shall be a sheet metal cover that is bolted to the frames after the plate pack has been assembled and pulled up to the proper dimension. The shroud shall be corrosion resistant, either by being constructed of corrosion-resistant material, such as CRES, or by being coated in corrosion-resistant material. It shall protect the plate heat exchanger assembly, including the gasketing, from damage and protect personnel from contact with hot plate edges and hot fluid in case of leaks. The shroud shall not extend below the bottom of the plate pack to

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ensure the emergence of any leakage into the drip pan is immediately noticeable. When necessary to meet fire testing requirements, the required shroud shall be included with the delivered PHE.

3.5.2 Inlet and outlet connections. Unless otherwise specified (see 6.2), inlet and outlet connections shall be flanged in accordance with ASME B16.5 at their outer end for connection to the piping system and project through the frame to contact the gasket on the first plate. Sacrificial liners may be utilized if approved by the Technical Authority (see 6.4). Materials shall be selected in accordance with MIL-STD-777. Note: Suggested materials are specified in table I.

3.5.2.1 Instrumentation and drains. Connections for pressure gauges, temperature instrumentation, high point vents, and low point drains shall be provided at inlets and outlets of cooled and cooling media as applicable. These connections shall be provided in the piping system between the heat exchange isolation valves and the heat exchanger. Associated piping materials shall be in accordance with MIL-STD-777 for the media intended.

3.5.2.2 Backflushing capability. Unless otherwise specified (see 6.2), for PHEs intended for use with seawater, the piping connecting to the PHE on the seawater side shall be arranged and valved to allow for backflushing the seawater side directly overboard for marine growth and debris removal.

3.5.3 Drip pan. A drip pan shall be incorporated in the design to contain and localize any plate leakage.

3.5.4 Connections. Connections to the ships' systems served shall be in accordance with ASME B16.5 and as specified (see 6.2).

3.5.5 Other aspects. The size and class of PHEs may entail additional construction aspects, including:

a. Carrying and support structures. When support feet, intermediate movable frames, carrying bars, guide bars, or other structural elements are necessary, they shall be made of the same materials as the frame.

b. Classes 3 and 4 HPEs. Plate-and-shell and welded block PHEs shall be of the same materials as gasketed PHEs, except for those components not utilized in those designs, such as tie bars.

3.6 Reliability and maintainability. Each PHE shall be designed for a mean time between failure of 50,000 hours with no unscheduled maintenance, where failure is defined as any conditions that prevents the PHE from meeting the performance characteristics in 3.7 or as otherwise specified (see 6.2). Scheduled maintenance, including plate replacement, or gasket renewal shall require no more than 2 hours per 20,000 hours of operation, without the use of special tools. Special tools are defined as tools not listed in the GSA Global Supply Catalog.

3.6.1 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.7 Performance characteristics. In addition to the design and construction characteristics herein, PHEs shall meet the performance characteristics of 3.7.1 through 3.7.5.

3.7.1 Operating pressure. Unless otherwise specified (see 6.2), the operating pressure capability of a PHE shall be up to 2.4 megapascal (mPa) (250 psi), for both sides of the heat transfer plates. Unless otherwise specified (see 6.2), the differential pressure between the heated media shall be a minimum of 35 kPa (5 psi) greater than that of the cooling media.

3.7.2 Operating temperatures. Unless otherwise specified (see 6.2), the operating temperature range of a PHE shall be up to 177 °C (250 °F), for both sides of the heat transfer plates.

3.7.3 Heat transfer load. The design heat transfer load shall be as specified (see 6.2). The specified load may be up to 5 megawatts (MW) (17,060,700 British Thermal Units per hour [Btu/hr]), with up to 1800 m^3/hr (8000 gallons per minute [gpm]) flow.

3.7.4 Leakage. Unless otherwise specified (see 6.2), there shall be no working fluid leakage from an assembled PHE. After shock loads, any leakage shall be immediately correctable by tightening the compression bolts (tie rods) for class 1 and 2 PHEs.

3.7.5 Fouling and wall shear stress. Unless otherwise specified (see 6.2), a 10 percent fouling margin shall be included and the minimum wall stress shall be 25 Pascals (Pa).

3.7.6 Maximum pressure drop. Unless otherwise specified (see 6.2), the maximum pressure drop across an HPE unit on both the cooling and cooled sides shall not exceed 100 kPa (15 psi).

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3.7.7 Fire testing. When required (see 6.2), PHEs shall meet fire testing specified in 4.9.3.

3.8 Human factors. The human engineering considerations specified in MIL-STD-1472, including instrumentation readability, lifting provisions, insulation or protection of hot surfaces, and safety interfaces required therein, shall be met. Insulation blankets such as those specified in MIL-PRF-2818 may be utilized.

3.9 Dimensions and weight. Unless otherwise specified (see 6.2) and accounting for the 15 percent plate addition capacity (see 3.4.1) and the 10 percent fouling margin (see 3.7.5), dimensions and weight of PHEs and their maintenance envelope shall be the minimum necessary to achieve heat transfer requirements (see 3.7.3).

3.10 Identification plates. Each PHE shall bear an identification plate in accordance with type A, B, C, D, or L of MIL-DTL-15024 and MIL-P-15024/5. Design pressure entries shall not appear on identification plates. All identification plates shall be normal service plate. Provisions shall be made for the following information:

a. Name of unit (HEAT EXCHANGER, PLATE TYPE).

b. Class of plate heat exchanger (see 1.2).

c. Manufacturer's service part number.

d. National stock number (allow 20 spaces).

e. Name of manufacturer.

f. Contract or order number. (The contracting activity's contract or purchase order number shall be entered in this space and shall utilize the width of plate to allow maximum number of spaces.)

g. Blank space for Defense Contract Management Agency (DCMA) stamp.

h. Date of manufacture.

i. Serial number.

j. Maximum hydrostatic test pressure.

k. Four blank spaces for unit number. (This space will be used for numbering the stamping to be done by the shipyard, when required for shipboard reference purposes)

l. Designation "U.S.".

3.11 Painting. Unless otherwise specified (see 6.2), only the frame, shroud, or other non-wetted, non-working surfaces shall be painted. Heat transfer plates, carry and guide bars, tightening bolts, fasteners, inlet and outlet port liners, plated surfaces (e.g., zinc plated pressure plate feet), and gaskets shall not be painted. Painted surfaces shall be finished as follows:

a. Surface preparation prior to painting in accordance with SSPC-SP 10/NACE No. 2.

b. One primer coat of formula 150, followed by two topcoats of epoxy paint formula 151, in accordance with MIL-DTL-24441, MIL-DTL-24441/20, and MIL-DTL-24441/21 (average 3 mils dry film thickness per coat).

3.12 Workmanship. All parts of the plate heat exchanger, before and after painting, shall be clean and free of sand, rust, dirt, pits, sprues, scale, and other harmful extraneous material that might detract from the intended operation, function, or appearance of the equipment. Fasteners shall be fully engaged, and bolt heads and nuts shall be flat on washers or mating surface. Sharp corners and projections, which may cause injury to personnel or on which clothing may catch, shall be avoided.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on each base size or new design of PHEs offered when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the first article tests specified in table II.

4.3 Conformance inspection. Conformance inspection of every PHE offered to the Government shall include the examination of 4.5 and the conformance tests specified in table II.

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TABLE II. Examinations and tests.

Characteristic	Requirement	Verification	First article	Conformance
Materials	3.2 through 3.2.3 and 3.3.3	4.4	X	X
Shock	3.3.1	4.5	X	
Mechanical vibration	3.3.2	4.6	X	
Operating environment	3.3.3	4.7	X	
Design	3.4 through 3.4.2	4.4	X	X
Construction	3.5 through 3.5.5	4.4	X	X
Reliability and maintainability	3.6, 3.6.1	4.8	X	
Performance	3.7 through 3.7.6	4.9.1, 4.9.2	X	X (4.9.2 only)
Fire test	3.7.7	4.9.3	X (if required)	
Human factors	3.8	4.4	X	
Dimensions and weight	3.9	4.4	X	X
Identification plates	3.10	4.4	X	X
Painting and workmanship	3.11, 3.12	4.4	X	X

4.4 Visual, dimensional, and materials examination. Each PHE offered for delivery shall be visually and dimensionally examined for materials, prohibited materials, welding and brazing, design, construction, human factors, dimensions, weight, identification plates, painting, and workmanship. Both the assembled heat exchanger and its plates shall be examined to the requirements listed in table II.

4.5 Shock test. Unless otherwise specified (see 6.2), shock tests shall be conducted in accordance with the requirements of MIL-DTL-901 for grade A equipment. Testing shall be conducted on one assembled and operating PHE.

4.6 Mechanical vibration. The PHEs shall be vibration tested in accordance with MIL-STD-167-1, type I.

4.7 Operating environment. All tests shall be conducted in a shop environment at temperatures between 4 to 71 °C (39 to 160 °F) and humidity between 70 and 95 percent.

4.8 Reliability and maintainability. The requirements of 3.6 shall be verified. The reliability of the PHEs shall be demonstrated by the successful completion of all tests and a reliability prediction. The prediction guidance provided in MIL-HDBK-217 may be used. The applicable maintenance procedures, such as replacement of plates, repairs, and the time required for accomplishment shall be verified during first article testing. The demonstration procedures in MIL-HDBK-470 may be used.

4.9 Performance. Performance capabilities required herein shall be verified by testing or analysis as specified.

4.9.1 Hydrostatic test. Each fluid side of a PHE shall be given a separate hydrostatic test at 1½ times the specified design pressure, using fresh water. The side not under hydrostatic test shall be open to atmospheric pressure. Test pressures shall be held for not less than 30 minutes, and leakage shall not occur.

4.9.2 Performance test. The heat transfer rate and pressure drops shall be determined by testing or analysis based on the intended fluid. Failure to meet heat transfer or maximum pressure drop requirements shall be test failures.

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4.9.3 Fire test. When specified (see 6.2) a fire test shall be conducted. A first article fire test shall be performed for the largest size plate heat exchanger used in petroleum to water applications. The plate heat exchanger shall be subjected to a fire test for 30 minutes for the purpose of verifying there is no hydrocarbon fluid leakage during the test. The heat exchanger fluids shall be maintained at design pressure during the test. The fire shall attain an average cold wall heat flux of 120 kilowatts per square meter (kW/m^2) within 5 minutes to all portions of the exterior surfaces of the heat exchanger and maintain that flux throughout the remainder of the test. If a fire shroud is required to meet these requirements, it shall be included with the delivered PHE.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The PHEs covered by this specification are intended for heat transfer equipment on naval ships.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class of plate heat exchanger (see 1.2).
c. Cooling media if not as specified (see 1.3.1.1, 1.3.1.2, 1.3.1.3, and 3.2).
d. When first article is required (see 3.1).
e. Shock requirements if other than specified (see 3.3.1).
f. Operating environment is other than specified (see 3.3.3).
g. Design aspects if other than specified (see 3.4.1).
h. Working fluids intended to be utilized (see 3.4.2).
i. Inlet and outlet flange connection, instrumentation, and backflushing requirements (see 3.5.2 and 3.5.2.2).
j. When attained reliability and maintainability metrics may be other than specified (see 3.6).
k. Operating pressures and differential pressure if other than specified (see 3.7.1).
l. Operating temperatures (see 3.7.2).
m. Heat transfer load requirements (see 3.7.3).
n. Allowable leakage, if not as specified (see 3.7.4).
o. If fouling margin and minimum wall stress are other than specified (see 3.7.5).
p. Maximum pressure drop if not as specified (see 3.7.6).
q. If fire testing is required (see 3.7.7 and 4.9.3).
r. Dimensions and weight (see 3.9).
s. If painted surfaces are other than specified (see 3.11).
t. If shock tests are other than specified (see 4.5).
u. Packaging requirements (see 5.1).
v. Provisioning requirements (see 6.5).
w. Recommended documentation (see 6.5).

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6.3 First article inspection. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, samples selected from the first lot size of production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05T may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.5 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition of any PHE under this specification (see 6.2):

a. Engineering drawings.
b. Materials listing.
c. Installation drawings.
d. Technical manual.
e. Test reports.
f. Certifications.
g. Onboard repair parts list and provisioning requirements.
h. Design report.
i. Heat transfer data sheets.

6.6 Subject term (key word) listing.

Cooling
Flat plate
Heat transfer
Heating
Working fluid

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CONCLUDING MATERIAL

Custodian:
Army –AV

Preparing activity:
Navy – SH
(Project 4420-2026-002)

Review activities:
Army – MI
DLA – CC

Civil agency:
GSA – FAS

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